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LLMs Get Lost In Multi-Turn Conversation

All Frontier Models Suffer a 39% Average Performance Drop Moving from Single-Turn to Multi-Turn Settings

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Every frontier language model tested—open- and closed-weight—performs significantly worse in multi-turn conversations than in equivalent single-turn tasks. Across more than 200,000 simulated conversations spanning six generation tasks, the average performance drop is **39%**. More troublingly, once a model takes a wrong conversational turn, it fails to recover: it gets lost and stays lost.

AT A GLANCE

Problem: LLMs are evaluated almost exclusively in single-turn, fully-specified instruction settings, masking a critical failure mode in multi-turn dialogue.

Why it matters: Real-world use involves extended conversations where user intent is underspecified and refined iteratively; models that degrade are unsafe to deploy in agentic contexts.

Method: 200,000+ simulated conversations on six generation tasks, decomposing the performance gap into aptitude loss and misalignment loss.

Result: 39% average drop across all frontier models; recovery after a wrong turn is essentially zero across all settings.

Evidence: GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, Llama 3.1 (70B/405B), Mistral Large all exhibit the same pattern.

The Big Picture

LLMs are conversational interfaces by design, yet nearly every benchmark measures performance on single isolated turns with fully specified prompts. This gap matters: analysis of real user logs shows that underspecification is the norm, not the exception. Users iterate, refine, and clarify. The question of how models perform across that iterative process has been largely ignored.

This paper closes that gap with a large-scale simulation study comparing single-turn and multi-turn performance on the same underlying tasks. The results are stark and consistent: every model tested degrades substantially in the multi-turn setting, and none recovers once a conversational wrong turn has been taken.

Why Existing Approaches Fall Short

Standard benchmarks like MMLU, HumanEval, and MT-Bench evaluate models on isolated, well-formed prompts. Even "multi-turn" benchmarks typically measure whether a model can carry context forward, not whether it can integrate iterative user refinements without losing the original intent.

The underspecification problem is orthogonal to context length: even short conversations (3–5 turns) show significant degradation. This rules out context-length limitations as the primary cause and points instead to a failure in *intent tracking* as conversation state evolves.

Core Mechanism

The authors decompose the multi-turn performance drop into two components:

Aptitude loss: A minor degradation in the model's core task ability when presented with conversation history rather than a direct prompt—small but measurable ($\approx$ 5–8 percentage points).

Misalignment loss: The dominant effect ($\approx$ 31–35 percentage points on average). As the conversation progresses, the model increasingly anchors to its own prior outputs rather than the user's current intent. A single wrong inference early in the conversation corrupts all downstream turns—the "getting lost" phenomenon.

Non-recovery: The recovery rate R_n—the probability that performance at turn $n + k$ exceeds performance at turn n after a detected misalignment—is approximately zero across all models, tasks, and conversation lengths tested.

Technical Formulation

Let $T = (t_1, t_2, \dots, t_n)$ be a conversation and y the target output. Model performance at turn n is:

$$\mathrm{Perf}_n = P(f(T_{1:n}) = y)\,.$$

The paper observes $\mathrm{Perf}_n < \mathrm{Perf}_1$ for all $n > 1$, all models, all tasks.

The *recovery rate* is defined as:

$$R_n = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[\mathrm{Perf}_{n+k} > \mathrm{Perf}_n]\,.$$

Empirically, $R_n \approx 0$ across all conditions.

The performance gap is decomposed as:

$$\Delta = \underbrace{(\mathrm{Perf}_1 - \mathrm{Perf}_1^{\mathrm{hist}})}_{\text{aptitude loss}} + \underbrace{(\mathrm{Perf}_1^{\mathrm{hist}} - \mathrm{Perf}_n)}_{\text{misalignment loss}}$$

where $\mathrm{Perf}_1^{\mathrm{hist}}$ is single-turn performance when history is present but the core prompt is unchanged.

Learning or Inference Procedure

The evaluation protocol creates multi-turn conversations by simulation:

1. Sample a task instance (x, y) from each benchmark.
2. Construct an underspecified initial prompt t_1 that omits key constraints or details.
3. Simulate a user who provides clarifications at each turn t_k, moving toward the fully-specified target.
4. Score the model's response at each turn against y.
5. Aggregate across 200,000+ conversations.

What the Guarantee Says

The paper does not provide formal guarantees, but its empirical finding is stated as a universal claim: *no evaluated model* achieves multi-turn performance within 20% of its single-turn performance, and *no evaluated model* has a recovery rate above 5% after a wrong turn.

Experimental Findings

Evaluated on six tasks—summarization, email drafting, code generation, QA, creative writing, data extraction—the paper reports:

- 39% average performance drop across all models and tasks.
- Code generation and QA: largest drops (≈45%).
- Creative writing: smallest drop (≈28%).
- Closed-weight models (GPT-4o, Claude, Gemini) do not outperform open-weight models (Llama, Mistral) on multi-turn robustness.
- Performance degradation begins at turn 2 and worsens monotonically through turn 5, then plateaus.

Ablations and Interpretation

Turn count: Degradation is sharpest between turns 1 and 2, then increases more slowly. The model's initial interpretation of the underspecified prompt is highly persistent.

Clarification type: Constraints added at turn 3+ are less reliably incorporated than those present at turn 1. Retroactive clarifications are especially poorly handled.

Model scale: Larger models within the same family show slightly smaller aptitude loss but equivalent misalignment loss — scale alone does not solve multi-turn robustness.

Implications: The result motivates dedicated multi-turn training objectives, retrieval-augmented intent tracking, and explicit user-intent summarization as architectural additions.

Reference

Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, Jennifer Neville. **LLMs Get Lost In Multi-Turn Conversation**. arXiv:2505.06120, May 2025. https://arxiv.org/abs/2505.06120